THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 10: **Light: Shadows and Reflections** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A 20 cm object is placed 40 cm from a lamp and its shadow falls on a screen at 100 cm from the lamp. The shadow height is: 1

(a) 40 cm **(b)** 50 cm **(c)** 100 cm **(d)** 20 cm

2. Which of these helps us to see a non-luminous book? 1

(a) the shadow of the book
(b) light passing through the book
(c) light from a lamp reflecting from the book
(d) the book giving its own light

3. A girl stands 2 m from a mirror and a wall is 5 m behind her. The image appears: 1

(a) 5 m behind the mirror **(b)** 7 m behind
(c) 2 m behind the mirror **(d)** on the wall

4. A stick 1 m high casts a shadow of 0.8 m. A tower casts a shadow of 40 m at the same time. The height of the tower is: 1

(a) 50 m **(b)** 80 m **(c)** 32 m **(d)** 40 m

5. Which surface gives a clear image of a tree? 1

(a) a woollen blanket **(b)** a still pond
(c) a rough road **(d)** a rough stone wall

6. A ray reflected by a plane mirror makes an angle of 100° with the incident ray. The angle of incidence is: 1

(a) 80° **(b)** 100° **(c)** 25° **(d)** 50°

7. A boy 1.5 m tall casts a shadow 2 m long. A tree casts a 12 m shadow. The tree height is: 1

(a) 12 m **(b)** 8 m **(c)** 16 m **(d)** 9 m

8. Why can we read a book by the reflected light from a lamp? 1

(a) paper reflects light diffusely **(b)** paper emits light
(c) paper is a mirror **(d)** paper is transparent

9. **Assertion (A):** A plane mirror forms a virtual image. 1
Reason (R): The image formed by a plane mirror can be obtained on a screen.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The image formed by a plane mirror is smaller than the object. 1
Reason (R): The image in a plane mirror is of the same size as the object.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Explain how a pinhole camera could be used to observe a solar eclipse safely. 2

12. Give a reason why a periscope's mirrors are plane. 2

13. A student at 3 m from a plane mirror moves to 5 m. By how much does the distance between her and her image change? 2

14. Why is the image formed by a plane mirror called virtual? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A 6 cm candle is placed 30 cm from a pinhole of a camera whose length is 15 cm. Find the image height. 3

16. Why does a polished steel plate show a clearer image than a rusty one? 3

17. How would you demonstrate lateral inversion with a piece of paper? 3

OR

A stick 1.2 m casts a shadow 0.9 m. At the same time a building casts a shadow of 30 m. Find the height. 3

18. Explain how a teacher can show that the image in a plane mirror is virtual. 3

OR

Explain why a shadow is not formed by a perfectly transparent glass but a clear glass may produce a faint shadow. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Divya stands 2 m in front of a large plane mirror. Behind an ambulance parked nearby, she reads the word written on its front as a mirror-reversed word and then understands why it is written that way. 5
(i) At what distance from the mirror is her image formed?
(ii) What is the distance between Divya and her image?
(iii) State two features of the image in a plane mirror.
(iv) Why is the word AMBULANCE written laterally inverted on the vehicle?

20. Explain, with the help of the laws of reflection and ray diagrams (in words), how a periscope gives an erect image and how the length of the tube may be changed. 5

OR

A circular disc of diameter 5 cm is placed 30 cm from a point source and a screen is 90 cm from the source. Find the diameter of the shadow, and the area ratio of shadow to disc. 5

END OF QUESTION PAPER — Total: 40 Marks